

# 语义表示与对比分析系统

## 1 Prompt

背景：在本周的学习中，我系统学习了文本的分布式表示方法。从基于矩阵分解的 LSA，到基于神经网络的 Word2Vec (CBOW/Skip-Gram)、GloVe，再到引入子词信息的 FastText 和句子级别的 Sent2Vec。我想开发一个集成多种语义表示模型的交互式 Web 系统。通过实际数据的运行，验证上述核心算法原理。

总体要求：使用 Python 的 Streamlit 框架，结合 gensim、scikit-learn 和 nltk 库，构建一个包含四个模块（多标签页结构）的 Web 应用。请准备一小段约 500-1000 词的英文文本（如维基百科语料、或文学作品等）作为实时训练的输入数据。四个模块解耦设计，设计四个 py 文件。

模块 1：传统统计模型 (TF-IDF 与 LSA)

在 Streamlit 应用的第一个标签页中，实现基于传统统计的文本表示。

- 1.提供一个文本域供用户输入英文语料，将其按句子切分为文档集合。
- 2.使用 `sklearn.feature_extraction.text` 计算并展示文本的 TF-IDF 矩阵，提取出权重最高的 5 个关键词。
- 3.在 TF-IDF 或 One-hot (CountVectorizer) 矩阵的基础上，应用 TruncatedSVD (LSA) 将词汇降维到 2 维空间。
- 4.使用散点图将降维后的词汇可视化在 2D 坐标系中。

在该模块中，我想观察基于共现矩阵分解的 LSA 方法，是否能将同现频率高的词语（如主语和相关动词）映射到相近的坐标。

模块 2：Word2Vec 训练与对比 (CBOW vs Skip-Gram)

在第二个标签页中，实现 Word2Vec 的实时训练和测试。

- 1.使用与模块 1 相同的语料（或其他语料），调用 `gensim.models.Word2Vec` 进行训练。
- 2.在界面上提供单选按钮，允许用户切换训练架构：CBOW (sg=0) 或 Skip-Gram (sg=1)。
- 3.提供滑动条 (Slider) 调整 window（上下文窗口，2-10）等参数。
- 4.提供一个单词输入框，计算并输出在当前训练模型下，与该词余弦相似度 (Cosine Similarity) 最高的 5 个词汇。

在该模块中，我想切换 CBOW 和 Skip-Gram 架构，观察同一个目标词的 Top 5 相似词是否发生变化

模块 3：预训练模型与词类比 (GloVe)

在第三个标签页中，测试预训练的 GloVe 模型与词类比任务。

1.在后台使用 `gensim.downloader` 加载预训练模型 `glove-twitter-25`（为保证速度可选用轻量级模型）。

2.构建词类比（Word Analogy）计算器：提供 A, B, C 三个输入框。

3.在后台计算向量公式： $\text{Result} = \text{Vector}(A) - \text{Vector}(B) + \text{Vector}(C)$ 。

4.检索并输出距离 Result 向量最接近的单词。

5.补充一个基础功能：输入任意两个单词，计算它们的词义相似度分数。

在该模块中，我想验证一些例子，A=king, B=man, C=woman，验证输出是否为 queen。尝试输入国家与首都（如 Paris - France + China），观察 GloVe 捕捉线性关系的能力

模块 4：子词特征与句向量 (FastText & Sent2Vec)

在第四个标签页中，实现 FastText 与句子级表示 (Sent2Vec)。

1.使用之前的语料实时训练一个 `gensim.models.FastText` 模型。

2.OOV 测试： 提供一个输入框，故意输入一个带有拼写错误的词（如 'computer'）。分别尝试用之前的 Word2Vec 和现在的 FastText 提取向量。用 try-except 捕获 Word2Vec 的 KeyError 并在界面提示“未登录词”，同时展示 FastText 成功计算出的最相似词。

3.Sent2Vec（简单实现）： 提供两个较长的句子输入框。将句子分词后，提取所有词的向量（可使用 FastText 向量）并求均值（Average Pooling），得到两个句子的全局向量（模拟课件 P81 的 Sent2Vec 基本思想）。

4.计算并输出这两个句向量的余弦相似度。

在该模块中，我想验证 FastText 利用 n-gram 字符级特征处理 OOV 问题时的鲁棒性；观察句向量均值法如何判断两个句子的全局语义相似度

调试 prompt:

为什么 king - man + woman 的结果最大可能性是 meet，而 queen 的概率仅为第三？

opencode 回答：这是因为 glove-twitter-25 是很小的模型，通常需要更大的模型（如 glove-wiki-gigaword-100）才能得到完美的 "queen" 结果

# 2 截图

## Interactive Web System for Semantic Representation Models

Welcome to this interactive demonstration of semantic representation models in Natural Language Processing. Navigate through the tabs below to explore different approaches to representing words and sentences as vectors.

### Learning Objectives:

- Understand TF-IDF and LSA for traditional text representation
- Compare CBOW and Skip-Gram architectures in Word2Vec
- Explore word analogies with pre-trained GloVe embeddings
- Test OOV handling with FastText's subword features

Module 1: TF-IDF & LSA    Module 2: Word2Vec    Module 3: GloVe & Analogy    Module 4: FastText & Sent2Vec

### Module 1: Traditional Statistical Models (TF-IDF & LSA)

This module demonstrates:

- TF-IDF (Term Frequency-Inverse Document Frequency) for keyword extraction
- LSA (Latent Semantic Analysis) using TruncatedSVD for dimensionality reduction

Goal: Observe how LSA maps co-occurring words (like subjects and verbs) to similar coordinates.

Enter English corpus (sentences will be treated as documents):

the-art performance on many NLP tasks. They are trained on massive amounts of text data and can capture complex linguistic patterns and relationships. Word embeddings are a type of word representation that allows words with similar meaning to have a similar representation. They are a key breakthrough in the field of natural language processing and have enabled significant improvements in various NLP tasks.

Natural language processing continues to evolve and improve, with new techniques and models being developed regularly. As more data becomes available and computational power increases, NLP systems are becoming more sophisticated and capable. The future of natural language processing holds great promise, with potential applications in healthcare, education, customer service, and many other domains. Understanding and processing human language is one of the most challenging and exciting areas of artificial intelligence research, and natural language processing is at the forefront of this exciting field.

Process Text

#### Step 1: Document Collection

Total documents (sentences): 40

> View documents

#### Step 2: TF-IDF Matrix

Documents

40

Vocabulary Size

100

Matrix Density

7.03%

TF-IDF Matrix (First 10 Docs × First 10 Terms)



#### Step 3: Top Keywords

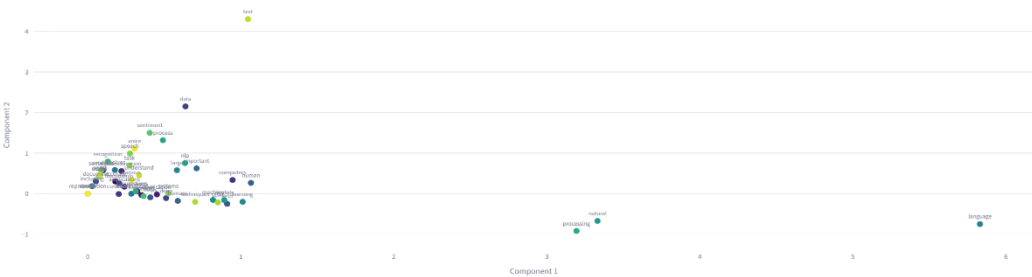
Top 5 Keywords by TF-IDF Score:

- language - Score: 5.0346
- text - Score: 3.7613
- natural - Score: 3.7058
- processing - Score: 3.3790
- data - Score: 2.7325

#### Step 4: LSA Word Embeddings (2D)

Words are embedded in 2D space using LSA. Words that frequently co-occur in similar contexts should appear close to each other in this visualization.

LSA Word Embeddings (2D Visualization)



Observation Tip: Look for semantically related words that cluster together. For example, words like 'language', 'processing', 'natural' might appear close to each other.

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## Module 2: Word2Vec Training and Comparison (CBOW vs Skip-Gram)

This module demonstrates:

- Real-time Word2Vec training on your corpus
- Comparison between **CBOW** (Continuous Bag of Words) and **Skip-Gram** architectures
- Effect of context window size on word embeddings

Goal: Compare the Top 5 similar words for the same target word under different architectures.

Enter English corpus for Word2Vec training:

Natural language processing is a machine learning technology that gives computers the ability to interpret, manipulate, and comprehend human language. Organizations today have large volumes of voice and text data from various communication channels like emails, text messages, social media newsfeeds, video, audio, and more. They use NLP software to automatically process this data, analyze the intent or sentiment in the message, and respond in real time to human communication.

Natural language processing helps computers communicate with humans in their own language and scales other language-related tasks. For example, NLP makes it possible for computers to read text, hear speech, interpret it, measure sentiment, and

Training Architecture: ☒ CBOW ☐ Skip-Gram

Content Window Size:

Vector Size:

Train Word2Vec Model

### Find Similar Words

Available words: and, the, language, text, natural, are, processing, data, nlp, with, that, sentiment, speech, human, many, models, important, for, process, recognition...

Enter a word to find similar words:

language

Top 5 words similar to 'language':

1. **power** - Cosine Similarity: 0.3355
2. **analyze** - Cosine Similarity: 0.2583
3. **today** - Cosine Similarity: 0.2437
4. **content** - Cosine Similarity: 0.2287
5. **communicate** - Cosine Similarity: 0.2195

### Word Embedding Visualization

Word2Vec Embeddings (2D PCA Projection)



### Architecture Comparison

Key Differences:

- **CBOW** (Continuous Bag of Words): Predicts the target word from context words
  - Faster training
  - Better for frequent words
  - Smoother representations
- **Skip-Gram**: Predicts context words from the target word
  - Slower training
  - Better for rare words
  - Captures more fine-grained semantics

Try This: Train both models and compare the Top 5 similar words for the same target word!

Compare Both Architectures

Enter word to compare:

nlp

Models trained! Showing comparison results:

CBOW Results

1. message (0.2657)
2. knowledge (0.2628)
3. generate (0.2561)
4. medical (0.2351)
5. for (0.2275)

Skip-Gram Results

1. message (0.2831)
2. knowledge (0.2766)
3. and (0.2681)
4. for (0.2670)
5. generate (0.2611)

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## Module 3: Pre-trained GloVe Model and Word Analogy

This module demonstrates:

- Pre-trained **GloVe** (Global Vectors for Word Representation) embeddings
- **Word Analogy** calculations using vector arithmetic
- Word similarity measurement

Goal: Verify classic examples like "king - man + woman = queen" and observe GloVe's ability to capture linear relationships.

✓ GloVe model loaded! Vocabulary size: 400,000

**Model Information:**

- **Model:** glove-wiki-gigaword-100 (100-dimensional vectors)
- **Training Data:** Wikipedia 2014 + Gigaword 5 (6B tokens)
- **Vocabulary:** ~400K words
- **Note:** This is a large, high-quality model trained on Wikipedia data. Classic word analogies like "king - man + woman = queen" should work correctly now!

**Word Analogy**   Word Similarity   Explore Words

### Word Analogy Calculator

Analogy Task: A is to B as C is to ?

Formula:  $Result = Vector(A) - Vector(B) + Vector(C)$

Classic Examples to try:

- A=king, B=man, C=woman → Expected: queen (may appear as top 3-5 due to model size)
- A=paris, B=france, C=italy → Expected: rome
- A=big, B=bigger, C=small → Expected: smaller

A (e.g., king)  
king

B (e.g., man)  
man

C (e.g., woman)  
woman

Number of results to show:  

10

Compute Analogy

king - man + woman = ?

✓ Expected word 'queen' found at rank 1!

**Top Results:**

1. queen - Similarity: 0.8524 (Expected)
2. throne - Similarity: 0.7664
3. prince - Similarity: 0.7592
4. daughter - Similarity: 0.7474
5. elizabeth - Similarity: 0.7460
6. princess - Similarity: 0.7425
7. kingdom - Similarity: 0.7337
8. monarch - Similarity: 0.7214
9. eldest - Similarity: 0.7185
10. widow - Similarity: 0.7099

**Word Analogy: king - man + woman = queen**

Legend:   
● king (A)   
● man (B)   
● woman (C)   
● queen (Result)

Note about results:

- This model uses 25-dimensional vectors trained on Twitter data
- Smaller models capture general semantic relationships but may not always return the "expected" answer first
- Related words (like 'prince' for royalty) may rank higher than the exact analogy

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## Module 4: Subword Features & Sentence Vectors (FastText & Sent2Vec)

This module demonstrates:

- FastText:** Handles OOV words using character n-grams
- Sent2Vec:** Sentence representation using average pooling of word vectors

Goals:

- Verify FastText's robustness with OOV words (e.g., misspellings)
- Observe how sentence similarity is computed using vector averaging

Enter English corpus for training:

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Train FastText Model

OOV Test    Sent2Vec

### Out-of-Vocabulary (OOV) Word Test

FastText uses character n-grams to handle words not seen during training. This allows it to find similar words for misspellings or new words!

Example: Try 'computer' (misspelling of 'computer')

Enter a misspelled or OOV word:

computeer

Correct spelling (for comparison, optional):

computer

Test OOV Handling

Word2Vec Result

✖ KeyError: Word not found in vocabulary

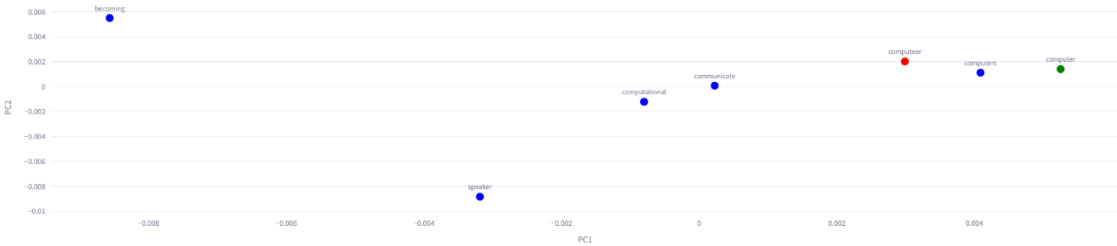
Word2Vec cannot handle OOV words!

FastText Result

✔ Word processed successfully!

- computers (0.5898)
- computational (0.4887)
- speaker (0.3467)
- becoming (0.3339)
- communicate (0.3245)

FastText OOV Handling: "computeer"



Why does FastText work?

FastText breaks words into character n-grams (e.g., "computer" → ["com", "omp", "mpu", "put", "ute", "ter"]). Even for OOV words like "computeer", it can construct a vector from shared n-grams with similar words!

**Key insight:** "computeer" shares n-grams like "com", "put", "ute" with "computer", allowing FastText to infer its meaning.

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Train FastText Model

OOV Test    **Sent2Vec**

### Sentence to Vector (Sent2Vec)

Simple Sent2Vec implementation: Average pooling of word vectors

Process:

1. Tokenize the sentence into words
2. Look up each word's vector
3. Compute the mean of all word vectors
4. The resulting vector represents the sentence

This captures the overall semantic meaning of the sentence.

Sentence 1:

Natural language processing is a machine learning technology.

Sentence 2:

Computers can understand human language using artificial intelligence.

Compute Sentence Similarity

**Sentence 1 Vector** (first 10 dimensions)

Shape: (100,)

[-0.0007, 0.0012, 0.0005, 0.0006, 0.0002, 0.0003, -0.0001, -0.0010, -0.0005, 0.0005...]

**Sentence 2 Vector** (first 10 dimensions)

Shape: (100,)

[-0.0003, -0.0001, -0.0005, -0.0000, 0.0003, -0.0001, 0.0012, 0.0004, -0.0003, -0.0008...]

### Similarity Result

Cosine Similarity: **0.0688**

Interpretation: Weakly similar sentences

### Word Breakdown

Sentence 1 words:

natural, language, processing, is, a, machine, learning, technology

Sentence 2 words:

computers, can, understand, human, language, using, artificial, intelligence

### 3 实验总结

在模块一中，使用 TF-IDF 提取文本关键词并应用 LSA 将高维词向量降至 2 维，观察到 TF-IDF 能够识别文档中的重要词汇、LSA 将共现频率高的词语映射到相近坐标，最后可视化展示了词语在潜在语义空间中的分布；模块二中使用了两种架构训练 Word2Vec 模型，CBOW：用上下文预测中心词，速度快，对常见词效果好、Skip-Gram：用中心词预测上下文，对稀有词更敏感，能捕捉更细粒度的语义，同一词汇在两种架构下的 Top 5 相似词会有一点不同；在模块三中使用 GloVe 通过全局共现统计捕捉词汇间的线性关系，模型维度越大，词类比效果越好；最后在模块四中，FastText 利用字符 n-gram 子词信息，输入拼写错误的词如 `computeer`，Word2Vec 报错，FastText 仍能找出相似词。Sent2Vec 通过平均池化将句子映射到向量空间，使用余弦相似度可衡量句子级别的语义相似性。